

Family 7.2: the global tensor with ocean colour, a real elevation static, and a build that reproduces recipe `f712` — what shipped, measured

blauewelt/earth — design note

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Abstract

Family 7.2 (build recipe `f712`, file stem `family7_global025_pentad_12`) is the published form of the programme’s global input tensor: the Earth’s surface and near-surface state on a 0.25° latitude–longitude grid from pole to pole, in fixed five-day bins from 1982 to the end of 2024, in 56 channels across four native-resolution groups, plus two static fields and two truth series. It supersedes two earlier builds. Family 7 (`f710`) carried three groups and two defects: an entire calendar year of observed sea-surface temperature and sea ice was absent, and two logarithmic channels were computed in a precision that does not reproduce. Family 7.1 (`f711`) added the fourth group — observed surface chlorophyll-a from the European Space Agency’s Ocean Colour Climate Change Initiative — and filled the missing year, but published an elevation static in which none of the 1,038,240 cells carried a number. Family 7.2 inherits three groups byte-for-byte from 7.1, rebuilds the fourth and both statics, and was afterwards rebuilt from the sources in an empty directory with nothing inherited: four of its five files came back byte-identical to the published bytes and the fifth is identical in every array value. This note states what is in the files, how to read them, what the per-channel coverage actually is, what the build refuses to do, and what it still cannot promise. A final section is a self-contained ingestion handover for an external, read-only agent.

1 Purpose, and what changed from 7 to 7.1 to 7.2

The programme builds a forward model of the Earth’s surface state: encode what is observed at each grid cell into a compact vector, predict that vector forward in time, and read out quantities of interest from the predicted state. The input to that model is a tensor, and the tensor described here is the third published generation of the global one. Its predecessors are not superseded by a change of science; they are superseded by three defects, each of which made the tensor either wrong or — more seriously — impossible to re-derive.

Family 7 (`f710`, 4 September 2026) is the first tensor in the programme to cover the whole planet rather than a North Atlantic window. Family 7.1 (`f711`, 14 September 2026) adds a fourth channel group, `oc025`, carrying the observed surface chlorophyll-a concentration from the European Space Agency’s Ocean Colour Climate Change Initiative (OC-CCI — a bias-corrected merge of the SeaWiFS, MERIS, MODIS-Aqua, VIIRS and OLCI satellite records) at 4 km, September 1997 to December 2024. Family 7.2 (`f712`, the same day) is the corrected build.

Defect 3, the one that is not about a number. Thirteen of the fifteen channels of the 1° atmosphere-and-land group are byte-identical across all three builds. The two that are not, `log_prate` and `log_swe`, are the only ones whose transform contains a transcendental function, and the builder was evaluating $\log(1+x)$ on a float32 array. Float32 $\log(1+x)$ is not correctly rounded: the result depends on the NumPy build and on which vector loop the processor dispatches to. Measured on a realistic precipitation sample, 8.1% of values differ by one or two units in the last place between the float32 loop and float64-then-cast, and about one in 22,000 of those flips the float16 that is finally stored. `f712` evaluates both channels in float64 and casts once, which is reproducible on any machine. The magnitude of the correction is scientifically nil — one float16 step is about 4×10^{-5} mm day $^{-1}$ of precipitation — and the property it buys is not: a tensor that does not reproduce cannot be re-derived and checked.

recipe	built (UTC)	groups	npz sha256	what it was and what was wrong
f710	2026-09-04 17:00:53, builder 56d07f4	g025, g100, rg100	27cd45eb...	the first global tensor. Defect 1: <code>sst</code> and <code>sea_ice</code> are entirely absent for bins 511–583, all of calendar 1989, because one source year could not be fetched and the stage marked the year done anyway. Defect 3 (below) present
f711	2026-09-14 15:19:34, builder 3a13f8a	+ oc025	91a8d86f...	adds ocean colour; all four groups computed from scratch, because the box holding the f710 work directory would not start. Fills 1989, so g025 legitimately differs from f710. Defect 2: <code>elev</code> published as 1,038,240 NaN. Defect 3 present
f712	2026-09-14 19:58:44, builder f25f1a6	same four	0837c450...	use this one. Inherits g025, rg100 and oc025 from f711 as hard links (same bytes, zero new storage); rebuilds g100 and both statics. Defects 2 and 3 closed

Table 1: The three published builds. Dates, builder commits and hashes are the `built_at`, `builder_git_sha` and `files[] .sha256` fields of each build’s own `manifest.json` on the public dataset repository. The f712 npz’s internal `built_at` reads 19:49:24 — the metadata stage ran nine minutes before the publish stage stamped the manifest.

Which recipe to use. f712. It is the only build whose elevation static carries data, the only one whose atmosphere group reproduces, and the only one that has been rebuilt from scratch and compared with itself (§9). f710 remains published as the historical original; f711 is superseded in every respect and is scheduled for deletion (§12).

2 Grid, time axis and bins

The horizontal grid is *point-aligned* at 0.25° : the grid points are the multiples of the step, with no half-cell offset, and the rows run *south first*. Latitudes are $-90 + 0.25i$ for $i = 0, \dots, 720$ (721 rows) and longitudes $-180 + 0.25j$ for $j = 0, \dots, 1439$ (1,440 columns), giving 1,038,240 cells; the arrays `lats` and `lons` in the small file state them. Longitude wraps: column 1439 (179.75° E) is adjacent to column 0. A coarse 1° grid of 181×360 points (`lat1`, `lon1`) is defined the same way, and a coarse channel is read at a fine position (y, x) by

$$y_1 = \min(\lfloor y/4 + \tfrac{1}{2} \rfloor, 180), \quad x_1 = \lfloor x/4 + \tfrac{1}{2} \rfloor \bmod 360, \quad (1)$$

so the two fine points on either side of a coarse point round to it and the last two columns wrap to -180° .

Time is a fixed five-day bin, a *pentad*, counted from the epoch 1982-01-01 (npz keys `epoch`, `cadence`, `pentad_days`): bin b covers days $[5b, 5b+5)$ after the epoch. There are 3,142 bins (`bin_index`), the first opening 1982-01-01 and the last, bin 3141, opening 2024-12-31 — which is why that bin holds one day of source data rather than five, and is empty or short in every group (§8). Bins are calendar-free: they do not align with months or years, and a year holds 73 of them. A bin is an averaging window, not a date.

Two groups do not have one row per bin. The monthly ocean-interior group `rg100`, built from the Argo array of profiling floats, has 252 rows, one per month from 2004-01 to 2024-12, each assigned to the pentad containing that month’s fifteenth; `rg_bin_index` gives the absolute bin of every row (1609 to 3137) and `rg_months` the month label. The colour group `oc025` has 1,997 rows on an *offset* axis: row r holds bin $1145 + r$, with `oc_bin_first` = 1145 the bin containing 1997-09-04, and `oc025_bin_index` listing 1145 to 3141 contiguously.

3 The four channel groups

Channels are stored in four groups, each an array of shape $[T, H, W, C]$ in little-endian float16, C (row-major) order, with a 128-byte NumPy header (measured on all four published files). The array is bin-major, so one time step of one group is a contiguous byte range (§7). Storing every channel on the fine grid instead would mean writing each 1.9° reanalysis value about sixty times.

group	grid	shape	bytes	content
g025	0.25°	[3142, 721, 1440, 7]	45,670,101,248	ocean surface
g100	1°	[3142, 181, 360, 15]	6,141,981,728	atmosphere and land, every surface
rg100	1° , monthly	[252, 181, 360, 32]	1,050,900,608	ocean interior, Argo
oc025	0.25° , from bin 1145	[1997, 721, 1440, 2]	8,293,461,248	ocean colour

Table 2: The four channel groups of **f712**, byte counts from **manifest.json**. With the 5,412,430-byte small file the published tensor is 61.16 GB, of which only **g100**’s 6.14 GB are bytes **f711** did not already hold.

Every channel is standardised before the float16 write, $x' = (x - \mu)/\sigma$, with (μ, σ) computed over all finite values in all bins and stored per group as **norm_<group>**. **A stored value is therefore in standard deviations, not in the channel’s unit.** The μ and σ columns of the four tables below are read directly out of those arrays; nothing is retyped. Float16 resolves a standardised value near 1 to about $10^{-3}\sigma$, which is 0.012°C for sea-surface temperature and $2 \times 10^{-4} \text{ m s}^{-1}$ for currents. Anomalies are not stored: the training code derives them at load time from a training-years-only climatology.

3.1 Ocean surface, g025

#	channel	unit	source	processing rule	μ	σ
0	cur_speed	m s^{-1}	GLORYS12 uo,vo	pentad mean of daily $\sqrt{u^2 + v^2}$; ≥ 3 days	0.1580	0.1594
1	log_mld	$\log_{10} \text{ m}$	GLORYS12 mldst	$\log_{10}$ of mixed-layer depth where > 0	1.4785	0.3587
2	ssh	m	GLORYS12 zos	pentad mean of sea-surface height above geoid	-0.1668	0.7129
3	cur_u	m s^{-1}	GLORYS12 uo	pentad mean, eastward; same pass as speed	-0.0007	0.1793
4	cur_v	m s^{-1}	GLORYS12 vo	pentad mean, northward	0.0059	0.1349
5	sst	$^\circ\text{C}$	OISST v2.1 sst	bilinear to point grid, pentad mean, ≥ 3 days; observed, never filled	13.6096	11.6202
6	sea_ice	fraction	OISST v2.1 icec	bilinear, pentad mean; absent below a concentration of 0.15	0.8267	0.2398

Table 3: Group **g025**. GLORYS12 is the Copernicus $1/12^\circ$ global ocean reanalysis [2]; OISST v2.1 is NOAA’s daily Optimum Interpolation Sea Surface Temperature analysis [3], an observation rather than a model. μ and σ are **norm_g025** from the published npz, in the channel’s own unit. **sst**’s constants differ from **f710**’s (13.615001, 11.622964) and **sea_ice**’s from (0.8271252, 0.23922408) because 1989 was added to the record before the statistics were taken.

Speed and the two velocity components are written in one pass, so $\sqrt{u^2 + v^2}$ equals the stored speed exactly. GLORYS12 covers 80°S northward and begins in 1993, so rows 0–39 of the five reanalysis channels are absent everywhere and bins 0–803 are absent entirely (§8). Sea-surface temperature is deliberately not filled over land or ice from any model.

3.2 Atmosphere and land, g100

The two soil channels are masked to land *before* regridding, so the coastal interpolation renormalises correctly, and are the only **g100** channels absent over sea. **skt**, the reanalysis skin temperature, is defined over every surface; it and **sst** are two different measurements by two different instruments and are stored as two channels rather than spliced at the coastline.

#	channel	unit	NCEP R1 file	processing rule	μ	σ
0	<code>tau_x</code>	N m^{-2}	<code>uflx.sfc</code>	sign-flipped: stress on the surface, eastward	0.0103	0.1246
1	<code>tau_y</code>	N m^{-2}	<code>vflx.sfc</code>	sign-flipped, northward	0.0042	0.0951
2	<code>tau_x_std</code>	N m^{-2}	<code>uflx.sfc</code>	within-pentad population σ of daily τ_x	0.0825	0.0724
3	<code>tau_y_std</code>	N m^{-2}	<code>vflx.sfc</code>	same, northward	0.0808	0.0721
4	<code>t2m</code>	$^{\circ}\text{C}$	<code>air.2m</code>	$\text{K} \rightarrow ^{\circ}\text{C}$	4.8070	21.9986
5	<code>u10</code>	m s^{-1}	<code>uwnd.10m</code>	eastward wind at 10 m	0.0027	4.3680
6	<code>v10</code>	m s^{-1}	<code>vwnd.10m</code>	northward wind at 10 m	0.1557	3.1669
7	<code>sp</code>	hPa	<code>pres.sfc</code>	$\text{Pa} \rightarrow \text{hPa}$	965.0467	94.7640
8	<code>log_prate</code>	$\log(1+x)$, mm d^{-1}	<code>prate.sfc</code>	$\text{kg m}^{-2} \text{s}^{-1} \times 86400$, then $\log(1+x)$ in float64	0.8662	0.7683
9	<code>log_swe</code>	$\log(1+x)$, mm	<code>weasd.sfc</code>	snow water equivalent, $\log(1+x)$ in float64	2.0701	3.5659
10	<code>soilw</code>	fraction	<code>soilw.0-10cm</code>	volumetric; land-masked before regridding	0.3040	0.0827
11	<code>tsoil</code>	$^{\circ}\text{C}$	<code>tmp.0-10cm</code>	$\text{K} \rightarrow ^{\circ}\text{C}$; land-masked before regridding	-9.6838	33.6608
12	<code>lhtfl</code>	W m^{-2}	<code>lhtfl.sfc</code>	latent heat flux, positive upward	62.3015	62.6118
13	<code>shtfl</code>	W m^{-2}	<code>shtfl.sfc</code>	sensible heat flux, positive upward	5.7186	40.1517
14	<code>skt</code>	$^{\circ}\text{C}$	<code>skt.sfc</code>	$\text{K} \rightarrow ^{\circ}\text{C}$; every surface, no mask, no splice	5.0877	22.7194

Table 4: Group `g100`, all from NCEP/NCAR Reanalysis 1 [4] — the 1948-onward global atmospheric reanalysis, on its T62 Gaussian grid of about 1.9° — bilinearly interpolated to the 1° point grid and averaged over the pentad with at least three days present. μ and σ are `norm_g100`, bit-identical across all three builds. The two channels marked float64 are the ones `f712` rebuilt; the reader should not expect that to move a mean taken over 2.82 billion values, and it did not.

3.3 Ocean interior, `rg100`

Thirty-two channels carry temperature and salinity at sixteen pressure levels from 10 to 1900 dbar, bilinearly interpolated from the product’s 1° cell-centred grid to the 1° point grid. The salinity standard deviation falls from 1.06 at 10 dbar to 0.13 at 1900 dbar, the widest scale range of any group.

3.4 Ocean colour, `oc025`

Two channels carry the value and the evidence it rests on. The group’s own time axis, block rule and measured coverage are §6; the table states what is stored.

4 Missing values

NaN — the floating-point “not a number” value — means “not observed here at this time” and is information rather than damage. The pattern is systematic. Ocean channels are absent over land; the GLORYS12 channels are absent south of 80°S and before 1993; soil channels are absent over sea; the Argo group is absent outside 2004–2024, outside its latitude band, and in any bin not on its monthly list; the colour group has no rows at all before bin 1145 — which is the period before any of those satellites flew, not a gap.

Two cases are easy to misread. **Sea-ice concentration is absent wherever it is below 15 %**, because that is how the source reports ice, so a consumer wanting zeros on open water sets the channel to zero where sea-surface temperature is finite and ice is not. And **`chl_cov` is NaN exactly where `log_chl` is, never 0**: a block that saw nothing has no coverage to report, and a stored zero would be indistinguishable from “observed, and the coverage was zero”. The tensor’s token design rests on that distinction.

A flat 56-channel view of a bin below 1145 has the two colour channels *absent* rather than NaN, and those are different states: absent means no satellite yet, NaN means the satellite looked and saw nothing. A consumer whose token design has no missing-versus-unobserved split should use NaN, which is the conservative choice.

Counting the finite values in the whole tensor by streaming every file (§8): 1.0757×10^{10} in `g025`,

level (dbar)	rg-t μ ($^{\circ}\text{C}$)	σ	rg-s μ (PSU)	σ	channel # (t, s)	finite cells/month
10	17.867	9.804	34.818	1.058	0, 16	32,611
30	17.519	9.679	34.864	1.032	1, 17	32,611
50	16.797	9.408	34.919	1.014	2, 18	32,611
100	14.731	8.363	35.004	0.961	3, 19	32,611
150	13.032	7.113	35.013	0.867	4, 20	32,611
200	11.704	6.022	34.962	0.777	5, 21	32,611
300	9.825	4.693	34.838	0.661	6, 22	32,611
400	8.436	3.884	34.739	0.595	7, 23	32,611
500	7.340	3.339	34.668	0.551	8, 24	32,611
700	5.792	2.609	34.604	0.488	9, 25	32,503
900	4.704	2.082	34.603	0.452	10, 26	32,503
1100	4.011	1.694	34.628	0.433	11, 27	31,621
1300	3.408	1.235	34.639	0.219	12, 28	31,205
1500	3.005	1.030	34.672	0.179	13, 29	31,205
1700	2.689	0.888	34.697	0.149	14, 30	31,205
1900	2.445	0.806	34.714	0.130	15, 31	31,205

Table 5: Group **rg100**: the Roemmich–Gilson Argo climatology [5] — an objective analysis of the global Argo profiling-float array — as temperature (**rg-t**, channels 0–15) and salinity (**rg-s**, channels 16–31) at sixteen pressure levels. μ and σ are **norm_rg100**. The last column is measured (§8): identical in every one of the 252 months and identical between temperature and salinity, and it falls with depth because the ocean’s floor rises into the deeper levels. Absent outside roughly 64.5°S – 79.5°N rather than extrapolated toward the poles.

#	channel	unit	source	processing rule	μ	σ
0	log_chl	$\log_{10} \text{mg m}^{-3}$	OC-CCI v6.0 chlora	per day, the mean of $\log_{10}$ chlorophyll over the finite cells of the 6×6 block; per pentad, the mean of those daily means over the days with at least one finite cell	−0.7765	0.4566
1	chl_cov	fraction in (0, 1]	OC-CCI v6.0 chlora	finite cell-days in the block during the pentad, divided by (cells in the block $\times$ days in the bin)	0.2823	0.2137

Table 6: Group **oc025**. μ and σ are **norm_oc025**. Full inversion of a stored value: a stored z becomes $\ell = z \times 0.4566048 - 0.7764690$ and the concentration is $10^{\ell} \text{mg m}^{-3}$; a stored z of zero is 0.167mg m^{-3} . Section 6 describes the group.

2.8169×10^9 in **g100**, 2.5953×10^8 in **rg100** and 1.7115×10^9 in **oc025** — 1.554×10^{10} observed values in total. The middle two figures agree exactly with the counts the builder’s own normalisation stage reports, which is a check on the scan rather than a coincidence.

5 The statics

Two static fields ship inside the small file, both [721, 1440] and both complete: the manifest and the npz agree that **static_nfinite** is {"elev": 1038240, "sphere": 1038240}. That key exists because the previous build published an elevation static with no data in it and nothing anywhere said so.

sphere is an **int8** surface class per cell: 0 ocean (702,642 cells), 1 land (226,495), 2 ice sheet or glacier (107,074, from Natural Earth’s glaciated-area polygons [7]), 3 inland water (2,029, from its lake polygons). Ice takes precedence over ocean where both apply, because the surface an instrument sees on an ice shelf is ice. The histogram is identical in all three builds and the array is bit-identical across them.

elev is float32 elevation in metres from ETOPO 2022 [6], *averaged over the cell*, negative under the sea, from $-9,687.95 \text{m}$ to $+6,004.33 \text{m}$. The convention matters when reading the maximum: a quarter-degree box in the Himalaya is about $28 \times 25 \text{km}$ and averages a summit with its valleys, so the highest cell on Earth reads $6,004 \text{m}$ rather than Everest’s $8,849 \text{m}$. The array is bit-identical to **f710**’s, compared cell by cell.

Two truth series are carried as (bin, value) pairs: **truth_rapid**, the overturning transport of the Atlantic at 26.5°N in Sverdrups as measured by the RAPID moored array, which has watched that

section continuously since 2004 — 1,459 pentads from bin 1626 to bin 3084; and `truth_fc`, the Florida Current cable transport, 2,490 pentads from bin 15 to bin 3141. Both are Atlantic; they exist so that a global model is scored on the same instrument as its predecessors.

6 The ocean-colour group

Colour is the only routine global observation of the ocean biosphere, and it is one of the read-outs the programme has committed to. It enters as a *target* and probe before it enters as an input: the question the group is built to answer is whether the physical state the model already carries — temperature, mixed layer, currents, wind, light through the calendar — predicts the biosphere a season ahead.

The block. A point of the 0.25° grid sits at a multiple of $0.25^\circ = 6/24^\circ$, which is a cell boundary of the source’s $1/24^\circ$ grid; the six source cells whose centres lie within $\pm 0.125^\circ$ of the point on each axis are therefore exactly three on either side. The block wraps in longitude; at the two pole rows it is truncated to the eighteen cells that exist, and the coverage denominator counts the cells that exist, so a fully observed pole block still reads 1.0.

Log space. Chlorophyll is log-normal over four orders of magnitude. The arithmetic mean of a block containing one bloom cell at 30 mg m^{-3} and thirty-five at 0.05 describes neither; the mean of $\log_{10}$ does, and the programme’s read-out for colour is scored in log space, so the stored quantity and the scored quantity agree.

What the built group measures. The design note for family 7.1 stated the rule and could not state the outcome; these are the numbers from the published bytes and from the npz’s own counter keys.

quantity	value	where it comes from
rows on the offset axis	1,997 (bins 1145–3141)	shape; <code>oc_bin_first</code> , <code>oc025_bin_index</code>
daily 4 km fields read	9,955	npz <code>n_occci_days</code>
days the archive listings did not offer	26	npz <code>n_occci_absent</code>
finite values per channel	855,742,985	npz <code>count_oc025</code> ; 41.3 % of $1997 \times 721 \times 1440$
colour on inland land cells	4,588,447 cell-pentads	npz <code>n_oc_inland</code> — measured, not masked
bin 2411 (2015-01-03): finite cells	421,137 of 1,038,240 (40.6 %)	E-077 RESULT 1, from the published bytes
bin 2411: global mean <code>log_chl</code>	−0.8295 (= 0.148 mg m^{-3})	the same
bin 2411: mean <code>chl_cov</code>	0.249	the same
bin 2411: <code>log_chl</code> range	−2.24 to +1.98 (0.006 to 95 mg m^{-3})	the same

Table 7: The colour group as built. Bin 2411 is the comparison pentad the family-7 validation already used, which is why it carries the eyeball check against the chlorophyll layers on the programme’s globe.

The design note for 7.1 predicted 1,997 rows from a first bin of 1145 and that is what shipped. **Colour over land is real and was left in:** the 4 km product resolves estuaries, lagoons and shelf water that the 0.25° `sphere` mask calls land. Those values are not masked away; `n_oc_inland` counts the finite colour cell-pentads sitting on land cells that touch no sea, so a consumer who wants open ocean only can mask them with `sphere`.

Where the colour actually came from. 1997–2022 from the per-file v6.0 archive of the Centre for Environmental Data Analysis (CEDA), whose daily directory listing ends at 2022; 2023–2024 from Plymouth Marine Laboratory’s aggregate `CCI_ALL-v6.0-DAILY` dataset, subset one day at a time, because that laboratory serves no per-year 4 km file directory at all. Both paths yield the same 4 km field and are reduced by the same code. The uncertainty fields shipped with the product are not stored — an uncertainty of a merged algorithm is not an observation — and the gap-filled level-4 products are not used, because the target should be what was seen.

7 Layout, and reading one bin over HTTP

Each group file is a plain NumPy `.npz`: a 128-byte header followed by little-endian float16 data in C order. Because the array is bin-major, one time step of one group is one contiguous range, and the public dataset repository answers `HTTP Range`: requests with status 206. The arithmetic is

$$\text{offset} = 128 + (b - b_{\text{first}}) \cdot H \cdot W \cdot C \cdot 2, \quad \text{length} = H \cdot W \cdot C \cdot 2, \quad (2)$$

where b_{first} is 0 for `g025` and `g100`, 1145 for `oc025` (or, equivalently, the row given by `oc025_bin_index`), and for `rg100` the row is the position of b in `rg_bin_index` rather than an offset. The slab lengths are 14,535,360 bytes for `g025`, 1,954,800 for `g100`, 4,170,240 for `rg100` and 4,152,960 for `oc025`.

```
import numpy as np, urllib.request

STEM = "family7_global025_pentad_l2"
BASE = ("https://huggingface.co/datasets/chfrank/earth-tensors/"
        "resolve/main/tensors/" + STEM + "/")
SHAPE = {"g025": (721, 1440, 7), "g100": (181, 360, 15),
         "rg100": (181, 360, 32), "oc025": (721, 1440, 2)}
FIRST = {"g025": 0, "g100": 0, "oc025": 1145} # rg100: use rg_bin_index

def read_bin(group, b, header=128):
    H, W, C = SHAPE[group]
    slab = H * W * C * 2 # float16
    row = b - FIRST[group]
    assert row >= 0, "that bin has no row in this group"
    lo = header + row * slab
    req = urllib.request.Request(
        BASE + f"{STEM}_X_{group}.npz",
        headers={"Range": f"bytes={lo}-{lo + slab - 1}"})
    with urllib.request.urlopen(req) as r:
        assert r.status == 206, "the host ignored Range - do not read on"
        buf = r.read()
    return np.frombuffer(buf, "<f2").reshape(H, W, C).astype(np.float32)

m = np.load(STEM + ".npz", allow_pickle=True) # the 5.4 MB small file
b = (np.datetime64("2015-01-03") - np.datetime64("1982-01-01")).astype(int) // 5
frame = read_bin("g025", b) # -> 2411, 14.5 MB, HTTP 206
c = list(m["chan_g025"].index("sst"))
sst_C = frame[..., c] * m["norm_g025"][c, 1] + m["norm_g025"][c, 0] # degrees C
```

The assertion on status 206 is load-bearing: a 200 means the host ignored the range header and is about to send 45.67 GB. The programme's own web front end paints a global frame this way, one range read per pentad.

8 Per-channel coverage, measured

Every published group file was streamed bin by bin in a sandbox with 30 GB of disk — nothing stored — and the finite cells counted per channel per bin. The bytes scanned are `f711`'s, which for `g025`, `rg100` and `oc025` are `f712`'s own bytes; `g100` differs between the two builds by at most one float16 step in about ten cells of a bin, which cannot change a finite count, and was independently confirmed full in all 3,141 live bins. The result is that **every channel is complete over its own record, and every hole in the tensor is a hole in a source**.

The last bin is the calendar, not the build. Bin 3141 opens 2024-12-31 and the sources end that day, so it holds one day rather than five. In `g025` and `g100` it is empty in every channel, because both require at least three days in a pentad. In `oc025` — whose rule needs only one day with a retrieval — it is short: 241,579 finite cells against a record mean of 428,514. A consumer that wants only full pentads drops bin 3141; a consumer that does not should know which of the two states it is in.

group	full-bin count	what the scan found
g025	693,205–693,310 (GLORYS12), 703,902 (sst)	the five GLORYS12 channels are empty for bins 0–803 and full from bin 804 (1993-01-03) — the reanalysis begins 1993-01-01 and bin 803 has only two of its days. sst is full in every one of the 3,141 live bins. sea_ice ranges 119,813–164,694 with the season and is empty in bins 433–439 (1987-12-06 to 1988-01-09, the changeover from the Scanning Multichannel Microwave Radiometer to the Special Sensor Microwave/Imager, a documented gap in the source), 2846 (2020-12-17) and 3090 (2024-04-20) — two isolated source outages
g100	65,160; 24,876 for the two soil channels	all fifteen channels full in all 3,141 live bins, with no exception anywhere. 24,876 is the 1° land mask
rg100	31,205–32,611 by level	all 32 channels full in all 252 months; the per-level counts of the table in §3.3, identical in every month
oc025	428,514 mean, 534,264 maximum	row 8 (bin 1153, 1997-10-14) is empty — the satellite returned nothing that pentad — and rows 0, 1, 2 (10,351 / 49,532 / 125,254 cells: the record’s first fortnight) and 88 (bin 1233, 142,136) are low. Everything else is ordinary cloud cover

Table 8: The finite-count scan. Counts are cells per bin per channel out of 1,038,240 at 0.25° and 65,160 at 1°.

9 The build, its guards, and the reproducibility result

The tensor is built by one resumable job, stage by stage, with a specification digest per stage so that a changed stage discards only its own state. **f712**’s stages are **glorys**, **sst**, **ncep**, **rg**, **occci**, **static**, **truth**, **norm**, **meta**, **publish**, and an opt-in **verify**.

The silent-skip audit. Every defect in §1 has the same mechanism: a source could not be read, one warning went into a log nobody reads after a green tick, and the build carried on and published a tensor with a hole in it that nothing downstream could see. Worse, the marker that says “this item is done” was written anyway, so a resume trusted it and the hole became permanent. Six such sites were found and closed in one pass, and the rule they now all obey is that **a marker may only under-claim**: the stage does not mark an item it could not read, and refuses at the end rather than marking itself done. The six were the monthly ocean-reanalysis chunk loader (whose month list was a filesystem glob, so a month the mirror did not hold was never even named); the reanalysis land mask (whose absence sent the two soil channels into the tensor *unmasked* over 71% of the planet, finite, plausibly scaled and indistinguishable from a masked tensor); the statics stage’s ocean mask; the Argo extension cubes (a bare skip left a *live* monthly row carrying nothing but NaN, which tells a model the subsurface was observed); and two paths in the truth stage, including an empty truth file that passed both existing guards.

The refusal flags. A degraded tensor now has to be asked for by name, and each flag records what it gave up: **--allow-empty-statics** (the stage may write an all-NaN elevation; the metadata and publish stages still refuse it, so the flag lets the stage finish and does not let the tensor ship), **--allow-missing-years** (a fill stage may mark itself done over a source year, a monthly chunk or the land mask), **--allow-empty-rg** (a zero-row subsurface group), and **--allow-drift** (the verification stage may report a difference and still exit 0 — without it a difference is a failed job, and the report is written either way, so the finding is never lost). Two guards are unconditional: the elevation fetch floor is sized from the 933 MB file it guards (0.10 MB s^{−1} over a 180-second probe) rather than from the 45 GB streams it was originally written for, and the metadata and publish stages refuse an empty static independently of any marker.

Reproducibility. A seeded build proves nothing about the builder. A hard link shows that the file system copied an inode; it says nothing about the code that wrote the bytes, which is exactly how an all-NaN elevation shipped green through a re-publish that verified the restore of the file it had been handed. So the tensor was rebuilt with nothing inherited: an empty work directory, no seed, the stage list **glorys**, **sst**, **ncep**, **rg**, **occci**, **static**, **truth**, **norm**, **meta**, **verify** and no publish stage at all, on the same rented box, 3 h 23 m for about \$1.1 at \$0.31 h^{−1}.

stage	s	what it read
glorys	2,669	the 2,339 daily 0.25° ocean cubes from the project's own mirror
sst	1,479	43 years of OISST from the mirror of NOAA's Physical Sciences Laboratory
ncep	1,400	645 reanalysis files from the same mirror
rg	20	the Roemmich–Gilson cubes, seeded from a release asset
occci	176	the 28 published per-year colour partials, each checked against its own sha256 before folding. Zero bytes from CEDA
static	5,774	Natural Earth (11 s) and ETOPO 2022 — two attempts, the first dying on an incomplete read after 42 min, the second finishing in 54
truth	0	the two label series, from the repository
norm	465	nothing; four groups' statistics and z-scores recomputed from the arrays just written
meta	4.5	the npz, 41 keys
verify	40	the five published files' sha256, and the published npz key by key

Table 9: The unseeded rebuild, stage by stage (E-077 RESULT 3).

The verdict: four of the five files byte-identical — g025 1cb1e1af..., g100 79c8075e..., rg100 55c49f1b..., oc025 c50c79b5..., each equal to the published hash — **and the fifth identical in every array value**. The npz's sha256 differs for a reason that is not about the data: `np.savez` writes a ZIP archive and a ZIP stores a per-member timestamp, so two archives of identical arrays cannot hash equally. The verification stage therefore opens both and compares key by key: 41 against 41, none only-local, none only-published, and the three that differ are the build's own signature — `builder_git.sha`, `built_at` and `sources`. Every array is bit-identical, `sphere` and `elev` included, and so are all four `norm_*` arrays, recomputed here from scratch over the 1.554×10^{10} observed values of §4.

One prefix decided it. The colour stage found its 28 per-year partials under a folder belonging to the *previous* recipe. Had it looked only under its own, nothing would have refused — an empty partials listing is a designed fallback to streaming the source, not a missing input — and the run would have streamed about 400 GB from an archive the same job had just measured at 1.20 MB s^{-1} , which is 92.6 h of transfer against a 24 h job timeout. The build would have died inside the colour stage having burned a box-day and would never have reached verification at all.

10 Verification

Every item below was checked against the published bytes: the small file downloaded whole, the group files by HTTP range read, the manifest by its own URL.

1. `static_n_finite` reads `{"elev": 1038240, "sphere": 1038240}` in both the manifest and the npz, and `sources` carries the ETOPO and Natural Earth entries. The elevation fetch took three attempts and 4,025 s at a probed 2.99 MB s^{-1} ; two attempts died on an incomplete read, and the re-sized floor let the third finish instead of aborting it.
2. `sphere`'s histogram is f711's exactly — 702,642 / 226,495 / 107,074 / 2,029 — and the array is bit-identical to both predecessors'.
3. `same_as_base` is `true` for g025, rg100 and oc025, with each hash equal to f711's, and `false` for g100, whose `base_sha256` (ba59ed7c...) is recorded beside this build's own (79c8075e...); the top-level `inherited_groups` and `rebuilt_groups` lists agree with that.
4. **norm_g100 did not move, against the prediction that it would.** All four `norm_*` arrays are bit-identical to f711's. The arithmetic that makes the prediction wrong is the same arithmetic that made the defect invisible: the rebuild moves about ten float16 cells per bin out of the 2,816,949,312 observed values the statistic is taken over — four parts in a billion — which cannot move a float32 mean or standard deviation. A moved `norm_g100` would have meant something else had changed. In total only five npz keys differ from f711's: `recipe`, `builder_git.sha`, `built_at`, `sources` and `elev`, plus the new `static_n_finite`.
5. The g100 difference, range-read on five bins (200, 1500, 2411, 3000, 3141): thirteen of fifteen channels byte-identical to *both* predecessors in every bin; `log_prate` differing from f711 in 9, 9, 9, 12 and 0 cells of 65,160 and `log_swe` in 1, 0, 2, 0 and 0; every difference exactly one float16

step; no NaN-payload and no signed-zero differences; bin 3141 identical in every channel. Note the direction: `f710` and `f711` differ from each other in only 1–3 `log_prate` cells per bin, so the float64 correction moves about four times as many cells as the drift it removes. That is expected — the two float32 builds were both wandering near the correctly rounded value and this one is at it.

6. The derived static grids the programme’s front end reads were regenerated at recipe `f712`: the elevation grid carries 1,038,240 non-null values where it carried 1,038,240 nulls, and the surface-class grid is byte-unchanged. The browser test that would have noticed the empty grid — a non-zero and above-90 % non-null count on the committed file — was added and checked to fail on the old file and pass on the new one.

11 Limitations

Colour reproduces from the partials, and is untested from the source. The colour reduction takes a float32 $\log_{10}$ on a 4320×8640 raster about ten thousand times, deliberately: a float64 working copy of that raster is 300 MB per temporary. Float32 $\log_{10}$ is not correctly rounded, so this is the same class of defect as `g100`’s two channels, and it was recorded as a known limitation rather than repaired. The unseeded rebuild narrows it considerably: the float32 $\log_{10}$ runs *once*, on the hosted runner that reduces a calendar year into its partial, and the box’s fold over those stored accumulators is exact arithmetic. So the claim is not “the colour group does not reproduce” but the strictly smaller “a colour year *re-reduced from the daily 4 km files* on a different machine is not expected to reproduce” — a path nothing has yet exercised, because every build so far has folded the same 28 partials. A chunked float64 $\log_{10}$ would close it at roughly zero extra peak memory whenever the group is next rebuilt for another reason.

The atmosphere group differs from its predecessors at the last bit. `g100` is the correctly rounded build and the two before it are not, but any of the three is scientifically usable: one float16 step is about 4×10^{-5} mm day⁻¹ of precipitation and 1.7×10^{-3} of a snow-water-equivalent unit, in about ten of 65,160 cells per bin, in two of fifteen channels. A consumer that keys a cache on a file hash must refresh; a consumer that does not may keep the older file.

The elevation maximum is a box mean. 6,004.33 m is the highest *cell average* of ETOPO 2022 over a 0.25° box, not the highest point on Earth. Anything that compares this static against a summit elevation is comparing two different quantities.

Standing limitations inherited from family 7. The atmosphere and land channels are a 1.9° reanalysis, the coarsest source in the tensor, and a stand-in for ERA5, which needs an account nobody in the programme yet holds; the layout is designed so that replacing the source changes no shape. Subsurface ocean structure is a monthly objective analysis at sixteen levels, present only from 2004 and only inside the Argo latitude band. The colour averaging discards 35 of every 36 source pixels; `chl_cov` records the loss but does not undo it, and the channel is clear-sky and daylight only, so it loses most of its high-latitude bins and all of the polar winter. And the record is 43 years, of which colour covers 27: no source shortens that constraint, and it is the reason the programme’s models are small.

12 Handover for an external agent

This section is written for an agent ingesting the tensor into its own infrastructure and needs no other document.

THE RULE: read only. The public dataset repository `chfrank/earth-tensors` on Hugging Face is your only source. You read it over anonymous HTTPS under `resolve/main/`; HTTP Range is supported and cross-origin reads are open. **You must never write to that repository, and never to the `blauewelt/earth source repository`** — no upload, no commit, no pull request, no branch, no file of any kind, whatever the reason and however convenient it would be. Everything you need is readable. If something appears to require a write, it is out of scope: report it instead (below).

The five files. Base URL, to which each file name is appended:

https://huggingface.co/datasets/chfrank/earth-tensors/resolve/main/tensors/family7_global025_pentad_12/

file (family7_global025_pentad_12...)	bytes	sha256
.npz	5,412,430	0837c450eba9e2cda4fb29ea81530987d5a8b5814d7722813fd100bc5595233f
X_g025.npy	45,670,101,248	1cb1e1af4132e6acdec76a7341a49453b32f9b982eee09d3ad577c5dd147d293
X_g100.npy	6,141,981,728	79c8075eedf68529736fff6c1f4c3cff0fef091322971919c4e400b9ed825b06
X_rg100.npy	1,050,900,608	55c49f1bbc55f9e82204d5d4ff2751cab5625c62003debe56df8cb5f1acaf2d5
X_oc025.npy	8,293,461,248	c50c79b5760103cdda1233fe6d6bf776fa03caf732a59b24cd27bd6a5cdd1700

Table 10: 61.16 GB in total. The same list, with per-file `inherited`, `same_as_base` and `base_sha256`, is in the folder’s `manifest.json`.

Verify before you trust. Four checks, in order of cost.

1. `sha256` of every downloaded file equals the file table below. The manifest URL is the base URL plus `manifest.json`, and it carries the same list, so a mismatch tells you whether the file or this document is stale.
2. In the `npz`: 41 keys; `recipe` is "f712"; `groups` has four entries; `oc_bin_first` is 1145; the index `oc025_bin_index` has 1,997 entries and its last is 3141; and `n_sst_days` is 15,706.
3. `static_n_finite` parses to `{"elev": 1038240, "sphere": 1038240}` and agrees with the arrays: `np.isfinite(elev).sum() == elev.size == 1038240`, and `sphere`’s four classes count 702,642 / 226,495 / 107,074 / 2,029.
4. Two spot reads. `g025` bin 547 (July 1989): `sst` 703,902 finite cells, `sea_ice` 153,002 — in `f710` both are zero, and that is the difference you are ingesting. `oc025` row 1266 (bin 2411, 2015-01-03): 421,137 finite `log_chl` cells, global mean `log_chl` -0.8295 after un-z-scoring, mean `chl_cov` 0.249, and `isnan(chl_cov) == isnan(log_chl)` everywhere.

The minimal loader. Memory-map a local copy, or range-read from the Hub with the code of §7. Either way the decode is the same three steps: interpret as little-endian float16, cast to float32, then un-z-score with the group’s `norm`. Do not hold `g025` in memory: it is 45.67 GB stored and 91 GB as float32.

```
import json, numpy as np

m = np.load("family7_global025_pentad_12.npz", allow_pickle=True)
groups = list(m["groups"]) # g025, g100, rg100, oc025
chans = {g: list(m["chan_" + g]) for g in groups}
norm = {g: m["norm_" + g] for g in groups}
n_fin = json.loads(str(m["static_n_finite"]))
assert n_fin["elev"] == m["elev"].size == int(np.isfinite(m["elev"]).sum())

def row_of(group, b):
    """pentad bin -> row in this group's array, or None if it has none."""
    if group == "rg100": # 252 monthly rows
        idx = m["rg_bin_index"]
        i = int(np.searchsorted(idx, b))
        return i if i < len(idx) and idx[i] == b else None
    if group == "oc025": # offset axis
        r = b - int(m["oc_bin_first"]) # 1145
```

```

        return r if 0 <= r < 1997 else None          # None: before 1997-09-04
    return b                                          # g025, g100: row == bin

X = np.load("family7_global025_pentad_l2_X_oc025.npy", mmap_mode="r")
frame = np.asarray(X[row_of("oc025", 2411)], dtype=np.float32) # [721,1440,2]
c = chans["oc025"].index("log_chl")
log10_chl = frame[..., c] * norm["oc025"][c, 1] + norm["oc025"][c, 0]
mg_per_m3 = 10.0 ** log10_chl                      # NaN stays NaN

```

If you already hold an earlier build. The deltas are small and exact.

you hold	re-fetch	keep
f711	the 5.4 MB npz (its <code>elev</code> is now real, and <code>static.nfinite</code> is new); <code>X.g100.npy</code> <i>only</i> if you key caches by file hash	<code>g025</code> , <code>rg100</code> , <code>oc025</code> — hard links of your files, hash-identical; rename or symlink. Delete any elevation you patched in from f710: it is identical, so this is bookkeeping
f710	the npz, <code>g025</code> (1989 is filled and the two observed channels were re-z-scored), <code>g100</code> and <code>oc025</code> (new)	<code>rg100</code> only — byte-identical, hash 55c49f1b..., so any subsurface-derived cache stays valid

Do not mix builds within a group: f712’s `g025` must be decoded with f712’s `norm_g025`. If your infrastructure fits its own standardisation or climatology on training years, recompute it — 1989 exists now, and any per-year table or fold definition that skipped it should stop skipping it.

Deprecation. `family7_global025_pentad_l1` will be **deleted from the Hugging Face dataset repository on 2026-10-15**. It is superseded in every respect by f712, whose three inherited files are the same bytes; nothing in it is unique. `family7_global025_pentad_l0` **stays** as the published original, because its hashes are cited elsewhere — but it carries the 1989 hole and the non-reproducing atmosphere group, so it is a historical record rather than a recommendation. Pin f712 by its hashes and re-read the manifest before a re-ingest.

Reporting a problem. Open an issue on the source repository, github.com/blauewelt/earth — a read-only agent can still open one, and an issue is not a write to the dataset. Quote the file name, its sha256, the bin and the check that failed. If issues are unavailable to you, report to Chris.

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